DATA MINING II: DATA WAREHOUSING

Concepts, Evolution, and Architecture of Modern Decision-Support Systems

An Introduction to Data Warehousing Fundamentals

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Data and Information

Data

Raw, unprocessed facts and figures — numbers, text, dates, or symbols — that by themselves carry no context or meaning.

Information

Data that has been processed, organized, or structured so it carries meaning and value for the person receiving it.

From Data to Information

Data

↓

Processing

↓

Information

Meaning and value are added at each step — information supports decisions in ways raw data cannot.

Evolution of Database

From flat files to cloud-native, distributed data platforms

1960s	**File-Based Systems**	Data stored in separate flat files per application; high redundancy
1970s	**Hierarchical & Network Models**	Data organized in tree or graph structures; rigid navigation paths
1970s–80s	**Relational Databases**	E.F. Codd’s relational model; data organized into related tables (RDBMS)
1990s	**Data Warehousing**	Subject-oriented repositories built for analysis, pioneered by Bill Inmon
2000s	**Distributed & NoSQL**	Web-scale systems handling volume, variety, and velocity of Big Data
2010s–Now	**Cloud & Modern Platforms**	Elastic, cloud-native warehouses and lakehouses (Snowflake, BigQuery)

File Processing vs. Database Processing

File Processing

- Each application maintains its own separate data files
- High data redundancy across files
- Data isolation — difficult to share between applications
- Little to no centralized control or standards
- Inconsistent data when the same fact is updated in only one file

Database Processing

- Data is centralized and shared across applications via a DBMS
- Redundancy is minimized through shared, structured storage
- Data integrity enforced by constraints and relationships
- Centralized security, backup, and access control
- A single update is reflected consistently for all users

Definition of Data Warehouse

> *"A data warehouse is a subject-oriented, integrated, time-variant, and non-volatile collection of data in support of management's decision-making process."*
>
> — Bill Inmon, widely regarded as the "father of data warehousing"

In simpler terms:

A data warehouse is a large, central store of data pulled together from many different systems across an organization, structured specifically so managers and analysts can query and analyze it to support better decisions — rather than to run daily operations.

Purpose of Data Warehouse

Support Decision-Making

Provide managers and analysts with reliable data to guide strategic and tactical decisions

A Single Version of Truth

Give the whole organization one consistent, agreed-upon set of figures and definitions

Enable Historical Analysis

Retain years of history so trends, patterns, and forecasts can be analyzed over time

Improve Data Quality

Clean, validate, and standardize data as it is integrated from source systems

Separate Analytics from Operations

Keep heavy analytical queries from slowing down operational transaction systems

Enable Business Intelligence

Serve as the foundation for dashboards, reports, and self-service analytics

Basic Principles of Data Warehouse

The foundational rules that guide how a data warehouse is built and run

1	**Separation from Operations**	The warehouse is kept physically and logically separate from operational (OLTP) systems so analytical queries never impact daily transactions
2	**Integration of Sources**	Data is drawn from many heterogeneous source systems and reconciled into consistent formats, codes, and naming conventions
3	**Read-Optimized Design**	Structures are optimized for large-scale reads and aggregations rather than for frequent single-record updates
4	**Historical Retention**	Data is kept over long time horizons, preserving snapshots rather than overwriting prior values
5	**Single Version of Truth**	Common definitions and business rules ensure every department reports the same numbers

Characteristics of Data Warehouse

Inmon's four defining properties of a true data warehouse

Subject-Oriented

Organized around major business subjects (customers, sales, products) rather than around individual applications

Integrated

Data from disparate source systems is made consistent in naming, format, encoding, and units of measure

Time-Variant

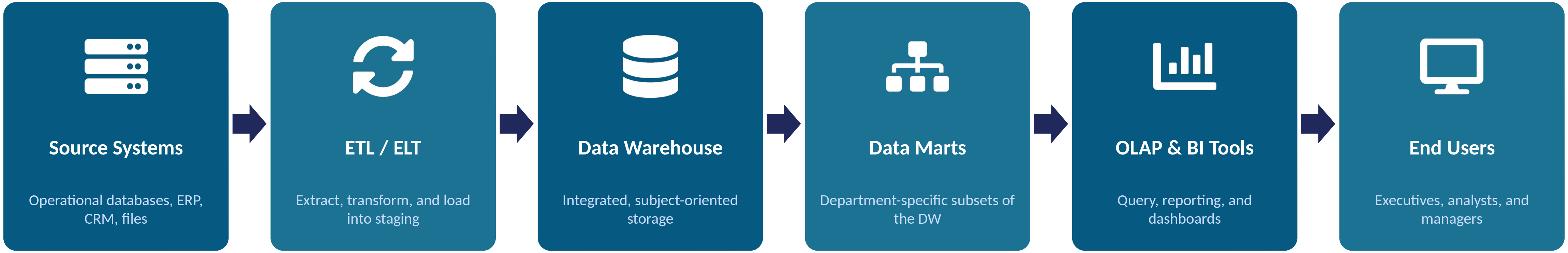
Data represents a series of snapshots over time, enabling trend and historical comparison

Non-Volatile

Once loaded, data is not updated or deleted in place — it is stable and read-only for users

Architecture of Data Warehouse

How data flows from source systems to end-user analysis



Metadata & Data Governance — spans every layer of the architecture

Two common architecture styles: a single-tier design loads a central warehouse directly for query, while a three-tier design adds a staging layer and data marts for scale and departmental flexibility.

Operational vs. Informational Systems

Operational Systems

(OLTP)

- Run the day-to-day business — orders, payments, inventory updates
- Process transactions in real time, one record at a time
- Hold current, highly detailed data
- Optimized for fast, frequent writes
- Used by front-line staff and applications

Informational Systems

(OLAP / Data Warehouse)

- Support management analysis and decision-making
- Process complex queries across large historical volumes
- Hold summarized and historical data
- Optimized for fast, complex reads
- Used by analysts, managers, and executives

Key Takeaway

From raw data to informed decisions — the data warehouse is
the bridge between daily operations and organizational insight.